Post-Pandemic Health Analysis: Patterns, Causes, and Insights

Joshitha Reddy B
IIT Bhilai
India

hyperref https://github.com/JoshithaBangaru/major-project

1 Data Overview and Preparation

A patient survey was conducted at Aadarana Clinic under the guidance of Dr. Chandhana Vanguru (NPT Neuro, Neuro Physical Therapist) and at City Neuro Hospital led by Dr. Vamshee Mohan Beeram (MS Neuro, Neurosurgeon).

The dataset consists of survey responses from individuals describing their health experiences during and after the pandemic.

1.1 Key Variables

- Age group, Gender
- Vaccination status (0–3 doses)
- Whether new health changes were observed
- Symptoms (multiple per individual)
- Timeline of symptoms (appearance, disappearance, reappearance)
- Lifestyle factors: Physical activity, Diet, Medical treatment
- Overall perceived health condition

1.2 Data Quality Challenges

- Same symptoms written in multiple ways (e.g., "hormonal", "hormonal imbalance")
- Spelling variations (e.g., "cholestrol", "cholesterol")
- Repeated words (e.g., "pain pain")
- Multiple symptoms combined in one field
- Missing or inconsistent separators

1.3 Data Cleaning Approach

- Symptoms were standardized and normalized
- Duplicate and repeated entries were removed
- Multi-symptom responses were converted into structured lists
- A rule-based classification approach was applied

2 Clustering

Clustering techniques (K-Means) were applied to group individuals based on symptom patterns.

This revealed natural groupings such as:

- High stress and fatigue group
- Physical pain group
- Low symptom group

Health changes after the pandemic are not random but follow identifiable patterns.

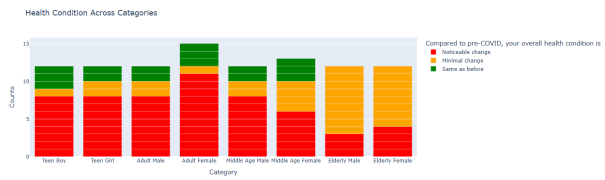

Figure 1: Health Distribution Across Categories

3 Executive Summary

What if the biggest impact of the pandemic wasn't just the virus… but everything around it?

A key observation is that many individuals reported new health issues that were not present before, including fatigue, stress, joint pain, and hormonal disturbances.

These changes were observed across different groups—regardless of whether they were infected or not—indicating broader impact.

Another important finding is that symptoms were often not stable. Many individuals reported that symptoms appeared after some time, disappeared temporarily, and then reappeared later.

Most importantly, individuals who maintained consistent self-care practices showed better recovery patterns.

3.1 Key Findings

- Many individuals reported new symptoms that did not exist before the pandemic
- Symptoms such as fatigue, stress, hormonal imbalance, and joint pain became common
- Some symptoms were temporary, disappeared, and later reappeared
- Vaccination showed mixed outcomes
- Individuals practicing consistent self-care showed better recovery and stability

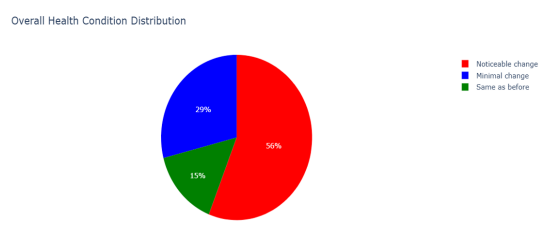

Figure 2: Overall Health Condition Distribution

4 Problem Framing

During the pandemic, three major forces acted together:

- The virus

- Vaccination
- Lifestyle disruption

The pandemic introduced sudden and prolonged changes in daily life:

- Reduced physical movement
- Increased stress and uncertainty
- Changes in eating habits and routines
- Greater dependence on screens and isolation

As a result, many individuals began to notice unexpected health changes.

A common concern among people has been:

- "I never had this issue before…"
- "This started after COVID or vaccination…"
- "It went away… but came back again…"

Hypothesis: Post-pandemic health issues are the result of a combination of physiological changes and lifestyle disruption.

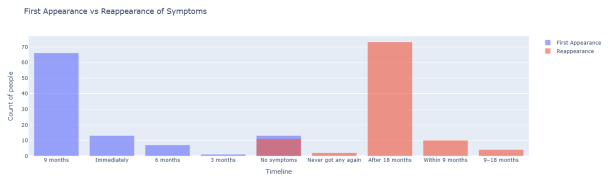

Figure 3: Symptom Appearance and Reappearance

5 Limitations of the Study

While this study provides meaningful insights into post-pandemic health patterns, it is important to acknowledge certain limitations.

5.1 Scope Restriction of Dataset

This dataset represents a general survey population, and the analysis was intentionally not restricted based on COVID infection status. Whether a person actually had COVID or not was not strictly verified or deeply analyzed.

5.2 Pre-Existing Health Conditions Not Considered

The dataset does not fully capture:

- Individuals' health condition before the pandemic
- Chronic illnesses or underlying conditions

Therefore, some symptoms may not be entirely new but could be continuations or aggravations of existing issues.

5.3 No Data on Severe Outcomes

The dataset does not include:

- Individuals who experienced severe complications
- Cases of death immediately after COVID or vaccination

The analysis focuses only on surviving and responding individuals, not extreme cases.

5.4 Heavy Data Cleaning Dependency

A major portion of this project involved:

- Cleaning inconsistent symptom data
- Standardizing names
- Removing duplicates

This introduces slight dependency on manual and rule-based assumption.

5.5 Vastness of the Problem

This topic is inherently very large and complex.

To fully analyze:

- Pandemic impact
- Vaccination effects
- Lifestyle changes
- Mental + physical health

We would need much larger datasets, clinical data, and long-term tracking.

6 Insights and Patterns

6.1 Dominance of Fatigue

Fatigue clearly stands out as the most widespread and consistently high-intensity symptom across almost all categories.

6.2 Strong Age-Dependent Shift in Symptom Type

Younger Population (Teens) Symptoms are mostly: Stress, Fatigue, Mild hormonal issues (especially in girls)

Insight: Younger individuals are less physically affected but more mentally impacted, likely due to academic pressure, screen time, and social disruption.

Adult Population Adult Females: High in hormonal imbalance, menstrual issues, and anxiety Adult Males: More balanced symptom distribution, with moderate fatigue and stress

Insight: Adults show a transition phase where both mental and physical symptoms begin to appear.

Middle-Aged Population Middle-Aged Males: Peak in stress, noticeable glucose-related issues Middle-Aged Females: Fatigue and moderate physical symptoms

Insight: This group is most vulnerable to lifestyle diseases and psychological stress.

Elderly Population Highest intensity in Joint pain, Fatigue, Decreased strength

Insight: Elderly individuals are primarily affected physically.

6.3 Mental Health Patterns

Figure 4: Mental Health Across Categories

6.4 Gender-Based Health Differences

Females: Hormonal imbalance, menstrual issues, fatigue **Males:** Stress, glucose issues, physical pain

6.5 Symptom Clustering Behavior

Cluster 1: Fatigue + Joint pain + Decreased strength **Cluster 2:** Stress + Glucose issues **Cluster 3:** Hormonal imbalance + Menstrual issues

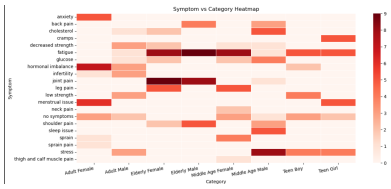

Figure 5: Symptom Heatmap

6.6 Vaccination Impact

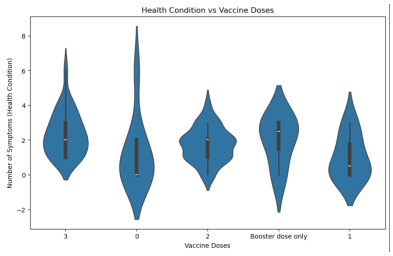

Figure 6: Vaccination vs Health Changes

6.7 Healthcare Access

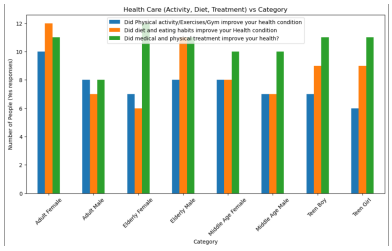

Figure 7: Healthcare and Recovery Patterns

6.8 Low Presence of "No Symptoms"

The "No symptoms" category appears across all groups but with low intensity.

6.9 Severity Concentration Zones

High-intensity regions are concentrated in elderly (physical symptoms) and middle-aged males (stress).

7 Story Arc

7.1 Context

2020: The world stopped. People stayed inside. Routines broke. Stress increased.

And slowly… People started noticing: "Something feels off."

7.2 Conflict

Not everyone was infected. Not everyone was sick.

Yet: new symptoms appeared old symptoms came back recovery was inconsistent

So the big question: What is actually happening to our health?

7.3 Resolution

We cleaned messy data. We classified symptoms. We clustered people.

And the pattern emerged:

Health issues are real They are widespread They are influenced by behavior Self-care changes the outcome

7.4 Impact

This changes the narrative completely:

The pandemic didn't just affect health—it changed how health behaves.

Health is now dynamic, not static Recovery is continuous, not instant Lifestyle is as important as medicine

8 Health Behavior Clustering using K-Means

8.1 Objective

The objective of this study is to group individuals based on their health-related lifestyle factors such as exercise, diet, treatment, and stress levels using unsupervised learning.

8.2 Dataset Description

The dataset contains survey responses collected during the COVID-19 period. It includes information about vaccination, symptoms, lifestyle habits, and health conditions.

8.3 Features Used

The following features were used for clustering:

- Exercise (Whether physical activity improved health)
- Diet (Whether diet improved health condition)
- Treatment (Whether medical treatment improved health)
- Stress level

All categorical variables were converted into binary values (Yes = 1, No = 0). Missing values were handled using simple imputation.

8.4 Methodology

K-Means clustering algorithm was applied with $k = 3$ to divide individuals into three behavioral groups.

Principal Component Analysis (PCA) was used to reduce dimensionality from 4 features to 2 components for visualization.

8.5 Results

The clustering algorithm successfully grouped individuals into three distinct categories:

- Cluster 0: Individuals with healthy lifestyle and low stress
- Cluster 1: Individuals with moderate lifestyle habits
- Cluster 2: Individuals with poor lifestyle and higher stress levels

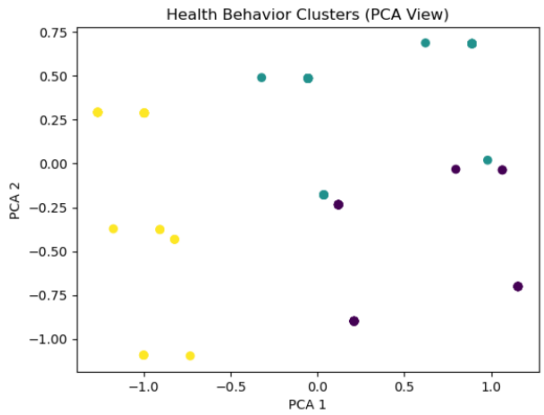

Figure 8: K-Means Clustering Visualization using PCA (2D Projection)

8.6 Conclusion

The study demonstrates that lifestyle factors significantly contribute to forming distinct health behavior patterns. K-Means clustering effectively identifies hidden patterns without labeled data.

9 Conclusion and Recommendations

Based on the analysis of post-pandemic health patterns, the following recommendations are proposed:

- Vaccines should be monitored for long-term side effects, while healthy individuals show better outcomes
- A disciplined lifestyle (exercise, diet, self-care) is essential for better recovery
- Regular health check-ups help in early detection and management
- Awareness of post-pandemic health effects should be increased
- Equal focus on physical and mental health is important